

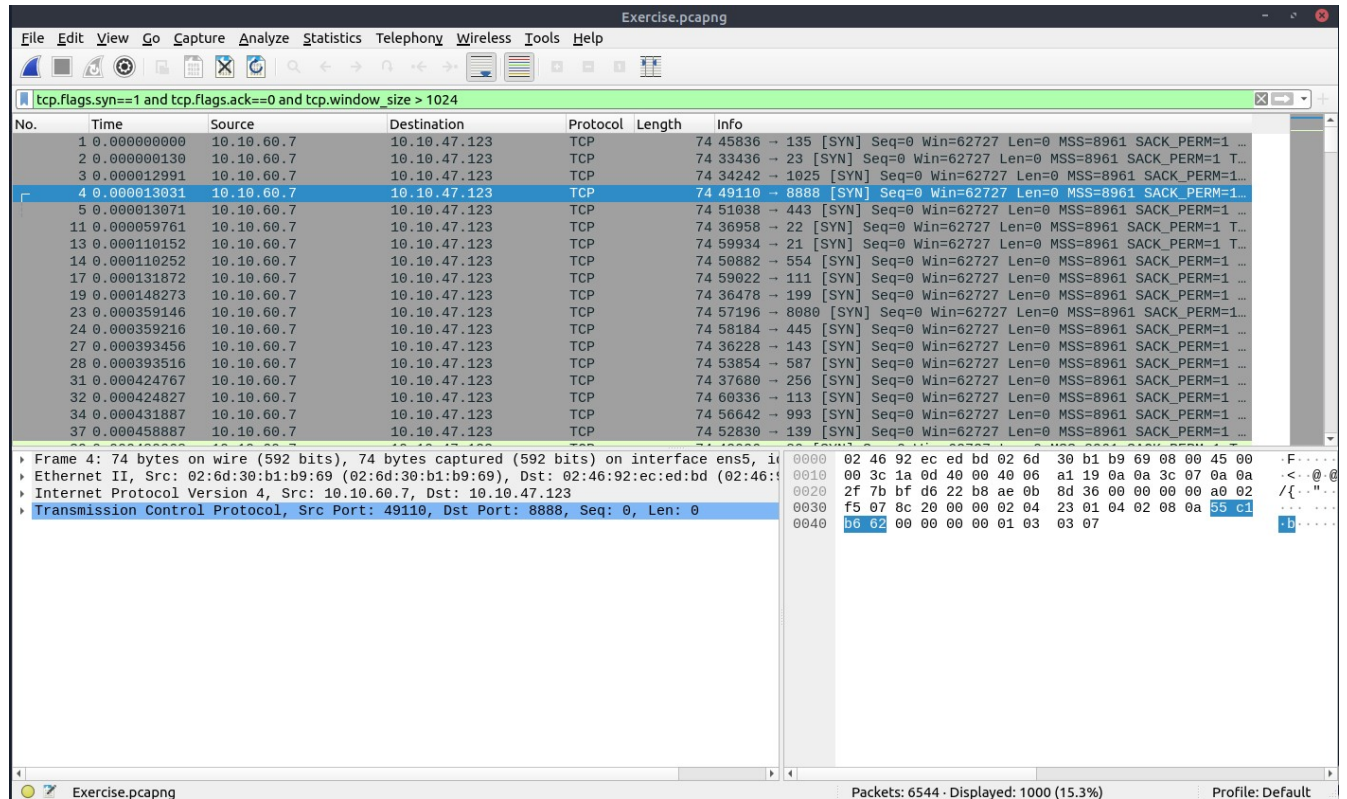
# Wireshark

## Traffic Analysis Workflow Example

### NMAP Scans

This is an example of how I researched a total number of TCP Connected scans that were greater than 1024 in a PCAP instance (~/Desktop/exercise-pcaps/nmap/Exercise.pcapng). A packet count of 1000 was displayed as the result using the following request/command:

`tcp.flags.syn==1 and tcp.flags.ack==0 and tcp.window_size > 1024`



The screenshot displays the Wireshark interface with a packet capture named 'Exercise.pcapng'. The packet list pane shows a filter: `tcp.flags.syn==1 and tcp.flags.ack==0 and tcp.window_size > 1024`. The list contains 37 packets, all of which are TCP SYN packets from source 10.10.60.7 to destination 10.10.47.123. The packet details pane shows the structure of a TCP SYN packet (Frame 4):

- Frame 4: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface ens5, in 0.000000000 seconds
- Ethernet II, Src: 02:6d:30:b1:b9:69 (02:6d:30:b1:b9:69), Dst: 02:46:92:ec:ed:bd (02:46:92:ec:ed:bd)
- Internet Protocol Version 4, Src: 10.10.60.7, Dst: 10.10.47.123
- Transmission Control Protocol, Src Port: 49110, Dst Port: 8888, Seq: 0, Len: 0

The packet bytes pane shows the raw data of the packet, with the following hex values highlighted in blue:

```
0000 02 46 92 ec ed bd 02 6d 30 b1 b9 69 08 00 45 00
0010 00 3c 1a 0d 40 00 40 06 a1 19 0a 0a 3c 07 0a 0a
0020 2f 7b bf d6 22 b8 ae 0b 8d 36 09 00 00 00 a0 02
0030 f5 07 8c 20 00 00 02 04 23 01 04 02 08 0a 55 c1
0040 b6 02 00 00 00 01 03 03 07
```

## ARP Poisoning/Spoofing (A.K.A Man-In-The-Middle Attack)

During this next scenario, I have opened a different PCAP file to investigate for an attack from a found MAC address `00:0c:29:e2:18:b4` and incorporated to the request/command: `arp.opcode == 1 && arp.src.hw_mac == 00:0c:29:e2:18:b4` to find the number of ARP requests crafted by the attacker which is 284 scans.

The screenshot shows a Wireshark capture of an ARP request. The filter bar at the top is set to `arp.opcode == 1 && arp.src.hw_mac == 00:0c:29:e2:18:b4`. The packet list shows 21 ARP requests from `VMware_e2:18:b4` to `Broadcast`. The packet details pane shows the structure of the Ethernet II frame and the ARP request.

| No. | Time         | Source          | Destination | Protocol | Length | Info                                     |
|-----|--------------|-----------------|-------------|----------|--------|------------------------------------------|
| 3   | 17.626854996 | VMware_e2:18:b4 | Broadcast   | ARP      | 42     | Who has 192.168.1.1? Tell 192.168.1.25   |
| 5   | 17.637345249 | VMware_e2:18:b4 | Broadcast   | ARP      | 42     | Who has 192.168.1.37? Tell 192.168.1.25  |
| 6   | 17.647731835 | VMware_e2:18:b4 | Broadcast   | ARP      | 42     | Who has 192.168.1.158? Tell 192.168.1.25 |
| 7   | 17.658130017 | VMware_e2:18:b4 | Broadcast   | ARP      | 42     | Who has 192.168.1.212? Tell 192.168.1.25 |
| 8   | 17.668703449 | VMware_e2:18:b4 | Broadcast   | ARP      | 42     | Who has 192.168.1.176? Tell 192.168.1.25 |
| 9   | 17.679601294 | VMware_e2:18:b4 | Broadcast   | ARP      | 42     | Who has 192.168.1.73? Tell 192.168.1.25  |
| 10  | 17.690243597 | VMware_e2:18:b4 | Broadcast   | ARP      | 42     | Who has 192.168.1.216? Tell 192.168.1.25 |
| 11  | 17.700760790 | VMware_e2:18:b4 | Broadcast   | ARP      | 42     | Who has 192.168.1.181? Tell 192.168.1.25 |
| 12  | 17.711256788 | VMware_e2:18:b4 | Broadcast   | ARP      | 42     | Who has 192.168.1.217? Tell 192.168.1.25 |
| 13  | 17.721858036 | VMware_e2:18:b4 | Broadcast   | ARP      | 42     | Who has 192.168.1.173? Tell 192.168.1.25 |
| 14  | 17.732272555 | VMware_e2:18:b4 | Broadcast   | ARP      | 42     | Who has 192.168.1.136? Tell 192.168.1.25 |
| 15  | 17.742493934 | VMware_e2:18:b4 | Broadcast   | ARP      | 42     | Who has 192.168.1.132? Tell 192.168.1.25 |
| 16  | 17.752775894 | VMware_e2:18:b4 | Broadcast   | ARP      | 42     | Who has 192.168.1.130? Tell 192.168.1.25 |
| 17  | 17.763563411 | VMware_e2:18:b4 | Broadcast   | ARP      | 42     | Who has 192.168.1.254? Tell 192.168.1.25 |
| 18  | 17.774149379 | VMware_e2:18:b4 | Broadcast   | ARP      | 42     | Who has 192.168.1.232? Tell 192.168.1.25 |
| 19  | 17.784781470 | VMware_e2:18:b4 | Broadcast   | ARP      | 42     | Who has 192.168.1.162? Tell 192.168.1.25 |
| 20  | 17.795271846 | VMware_e2:18:b4 | Broadcast   | ARP      | 42     | Who has 192.168.1.109? Tell 192.168.1.25 |
| 21  | 17.805791112 | VMware_e2:18:b4 | Broadcast   | ARP      | 42     | Who has 192.168.1.253? Tell 192.168.1.25 |

Frame 3: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface eth0, 192.168.1.25  
Ethernet II, Src: VMware\_e2:18:b4 (00:0c:29:e2:18:b4), Dst: Broadcast (ff:ff:ff:ff:ff:ff)  
Address Resolution Protocol (request)

I have then searched for the number of HTTP packets received by the attacker by using the instance `http && eth.dst == 00:0c:29:e2:18:b4` to find that the number of packets were 90

The screenshot shows a Wireshark capture of HTTP traffic. The filter bar at the top is set to `http && eth.dst == 00:0c:29:e2:18:b4`. The packet list shows 21 HTTP requests from `192.168.1.12` to `44.228.249.3`. The packet details pane shows the structure of the Ethernet II frame and the HTTP request.

| No.  | Time          | Source       | Destination  | Protocol | Length | Info                                                            |
|------|---------------|--------------|--------------|----------|--------|-----------------------------------------------------------------|
| 1107 | 107.294748849 | 192.168.1.12 | 44.228.249.3 | HTTP     | 509    | GET /login.php HTTP/1.1                                         |
| 1116 | 107.524664472 | 44.228.249.3 | 192.168.1.12 | HTTP     | 1350   | HTTP/1.1 200 OK (text/html)                                     |
| 1122 | 107.544502796 | 192.168.1.12 | 44.228.249.3 | HTTP     | 420    | GET /style.css HTTP/1.1                                         |
| 1123 | 107.546558129 | 192.168.1.12 | 44.228.249.3 | HTTP     | 461    | GET /images/logo.gif HTTP/1.1                                   |
| 1137 | 107.767662150 | 44.228.249.3 | 192.168.1.12 | HTTP     | 906    | HTTP/1.1 200 OK (GIF89a)                                        |
| 1143 | 107.768836318 | 44.228.249.3 | 192.168.1.12 | HTTP     | 1180   | HTTP/1.1 200 OK (text/css)                                      |
| 1167 | 107.856353223 | 192.168.1.12 | 44.228.249.3 | HTTP     | 457    | GET /favicon.ico HTTP/1.1                                       |
| 1217 | 108.088537576 | 44.228.249.3 | 192.168.1.12 | HTTP     | 948    | HTTP/1.1 200 OK (image/x-icon)                                  |
| 1226 | 112.878779336 | 192.168.1.12 | 44.228.249.3 | HTTP     | 711    | POST /userinfo.php HTTP/1.1 (application/x-www-form-urlencoded) |
| 1232 | 113.105809063 | 44.228.249.3 | 192.168.1.12 | HTTP     | 60     | HTTP/1.1 200 OK (text/html)                                     |
| 1272 | 128.722212965 | 192.168.1.12 | 44.228.249.3 | HTTP     | 825    | POST /userinfo.php HTTP/1.1 (application/x-www-form-urlencoded) |
| 1275 | 128.953703403 | 44.228.249.3 | 192.168.1.12 | HTTP     | 1488   | HTTP/1.1 200 OK (text/html)                                     |
| 1280 | 130.293866830 | 192.168.1.12 | 44.228.249.3 | HTTP     | 825    | POST /userinfo.php HTTP/1.1 (application/x-www-form-urlencoded) |
| 1283 | 130.519826760 | 44.228.249.3 | 192.168.1.12 | HTTP     | 1488   | HTTP/1.1 200 OK (text/html)                                     |
| 1288 | 134.047286876 | 192.168.1.12 | 44.228.249.3 | HTTP     | 587    | GET /logout.php HTTP/1.1                                        |
| 1291 | 134.271756279 | 44.228.249.3 | 192.168.1.12 | HTTP     | 1180   | HTTP/1.1 200 OK (text/html)                                     |
| 1298 | 136.609621027 | 192.168.1.12 | 44.228.249.3 | HTTP     | 557    | GET /login.php HTTP/1.1                                         |
| 1301 | 136.829973665 | 44.228.249.3 | 192.168.1.12 | HTTP     | 1350   | HTTP/1.1 200 OK (text/html)                                     |

Frame 1107: 509 bytes on wire (4072 bits), 509 bytes captured (4072 bits) on interface eth0, 192.168.1.12  
Ethernet II, Src: VMware\_98:c7:a8 (00:0c:29:98:c7:a8), Dst: VMware\_e2:18:b4 (00:0c:29:e2:18:b4)  
Internet Protocol Version 4, Src: 192.168.1.12, Dst: 44.228.249.3  
Transmission Control Protocol, Src Port: 49915, Dst Port: 80, Seq: 1, Ack: 1, Len: 455  
Hypertext Transfer Protocol

I then looked at the number of sniffed username & password entries by writing the instance `((eth.dst == 00:0c:29:e2:18:b4 and http) ) && (urlencoded-form.key == "pass")` only to find that 6 usernames and password entries were used in the attempts of sniffed packets.

The image shows a Wireshark capture of network traffic. The filter bar at the top displays the filter `((eth.dst == 00:0c:29:e2:18:b4 and http) ) && (urlencoded-form.key == "pass")`. The packet list pane shows six HTTP POST requests to `/userinfo.php` from source IP `192.168.1.12` to destination IP `44.228.249.3`. The selected packet (No. 1226) is expanded in the packet details pane, showing the following layers:

- Frame 1226: 711 bytes on wire (5688 bits), 711 bytes captured (5688 bits) on interface
- Ethernet II, Src: VMware\_98:c7:a8 (00:0c:29:98:c7:a8), Dst: VMware\_e2:18:b4 (00:0c:29:e2:18:b4)
- Internet Protocol Version 4, Src: 192.168.1.12, Dst: 44.228.249.3
- Transmission Control Protocol, Src Port: 49915, Dst Port: 80, Seq: 1225, Ack: 9605, Len: 711
- Hypertext Transfer Protocol
- HTML Form URL Encoded: application/x-www-form-urlencoded

The packet bytes pane shows the raw data of the selected packet, starting with the Ethernet II header and followed by the IP and TCP headers, and the HTTP POST body.